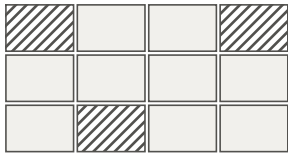


a

tiles of one section



 training tiles: fit the baseline and the probe

 held-out tiles: score both

baseline: $\bar{v}_k(t)$, the mean over cells of type t

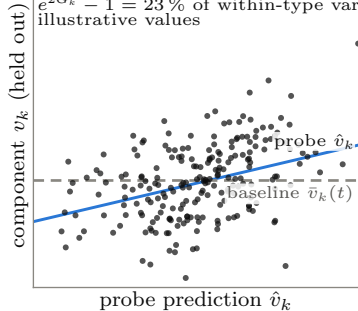
probe: $\hat{v}_k(\mu_z, t)$, ridge or MLP

b

$$G_k = \frac{1}{2} \log \frac{\text{MSE}_{\text{type}}}{\text{MSE}_{\text{probe}}} = 0.10 \text{ nats}$$

$$e^{2G_k} - 1 = 23\% \text{ of within-type variance}$$

illustrative values

**c**

 floor  this fit  uncontrolled
 guard: $\frac{1}{4}$ of uncontrolled
 composition



fraction of uncontrolled = 0.31: fails

image



fraction of uncontrolled = 0.16: passes

G_b (nats per component), illustrative